

Models:

1. LASSO - Baseline Model

=== STATIC LASSO — Walk-Forward Validation ===

Step 1 | Train:2010 → Eval:2015 AUC=0.5000 AP=0.0462 (best C=0.0010)

Step 2 | Train:2010+2015 → Eval:2020 AUC=0.5305 AP=0.1144 (best C=0.0728)

=== DYNAMIC LASSO — Walk-Forward Validation ===

Note: 2010 rows excluded (no prior period for lags/deltas)

Step 1 | Train:2015 → Eval:2020 AUC=0.4808 AP=0.0983 (best C=0.0067)

2. XGBoost - Feature engineering (Accounts for interactions)

=== STATIC XGBoost — Walk-Forward Validation ===

Step 1 | Train:2010 → Eval:2015 AUC=0.8250 AP=0.1778 (depth=4 lr=0.1 n=100)

Step 2 | Train:2010+2015 → Eval:2020 AUC=0.5366 AP=0.1253 (depth=4 lr=0.1 n=100)

=== DYNAMIC XGBoost — Walk-Forward Validation ===

Note: 2010 rows excluded (no prior period for lags/deltas)

Step 1 | Train:2015 → Eval:2020 AUC=0.8226 AP=0.2583 (depth=2 lr=0.05 n=100)

3. Random Forest

4. GEE (Generalized Estimating Equations)

- a. Acknowledges that the same tract appears in 2010 and 2015
- b. Corrects for within-tract correlation and gives you population-average coefficients that are more defensible automatically

Conditions for all models:

- Try both static and temporal feature sets
 - Temporal includes lagged values, first differences, and the AR term
- Gentrify_density -> combined density of arts + food service jobs per square mile

Walk-Forward Validation

	Static	Temporal
LASSO	AUC: 0.5305 AP: 0.1144	AUC: 0.4808 AP: 0.0983
XGBoost		
Random Forest		
GEE		

=== Full Model Comparison on 2020 Validation Set ===

Model	AUC	AP

Static LASSO	0.5305	0.1144
Dynamic LASSO	0.4808	0.0983
Static XGBoost	0.5366	0.1253
Dynamic XGBoost	0.8226	0.2583

Baseline (random) AP $\approx$ 0.0983

Gentrification is multi-dimensional. Predicting on all at once limits the dataset too much, one at a time is more accurate.

Running 3 Models to Predict:

- Appreciation: Captures investment pressure
- Rent Growth: Captures housing affordability loss
- Income Growth: Captures demographic replacement

Combine the three predicted values into a composite index -> Richer risk score

Results:

- Must remove 2010 data, since it captures the 2008 financial crisis which was very impactful on appreciation and home values; not generalizable at all

